Chapter 3

MATRICES

EXERCISE 3.1 (Page 126)

1. Let

$$A = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$

Find: (i) $A + B$

Sol. $$A + B = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} + \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2-1 & -3+3 & -5+5 \\ -1+1 & 4-3 & 5-5 \\ 1-1 & -3+3 & -4+3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

1.(ii) $A - B$

Sol. $$A - B = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} - \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} + \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2+1 & -3-3 & -5-5 \\ -1-1 & 4+3 & 5+5 \\ 1+1 & -3-3 & -4-3 \end{bmatrix} = \begin{bmatrix} 3 & -6 & -10 \\ -2 & 7 & 10 \\ 2 & -6 & -7 \end{bmatrix}$$

1.(iii) $2A + 3B$

Sol. $2A + 3B = 2\begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} + 3\begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$

$$= \begin{bmatrix} 4 & -6 & -10 \\ -2 & 8 & 10 \\ 2 & -6 & -8 \end{bmatrix} + \begin{bmatrix} -3 & 9 & 15 \\ 3 & -9 & -15 \\ -3 & 9 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 4-3 & -6+9 & -10+15 \\ -2+3 & 8-9 & 10-15 \\ 2-3 & -6+9 & -8+9 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \\ 1 & -1 & -5 \\ -1 & 3 & 1 \end{bmatrix}$$

1.(iv) $3A - 5B$

Sol. $3A - 5B = 3\begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} - 5\begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$

$$= \begin{bmatrix} 6 & -9 & -15 \\ -3 & 12 & 15 \\ 3 & -9 & -12 \end{bmatrix} + \begin{bmatrix} 5 & -15 & -25 \\ -5 & 15 & 25 \\ 5 & -15 & -15 \end{bmatrix}$$

$$= \begin{bmatrix} 6+5 & -9-15 & -15-25 \\ -3-5 & 12+15 & 15+25 \\ 3+5 & -9-15 & -12-15 \end{bmatrix} = \begin{bmatrix} 11 & -24 & -40 \\ -8 & 27 & 40 \\ 8 & -24 & -27 \end{bmatrix}$$

1.(v) AB

Sol.

$$AB = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix}\begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2\times(-1)-3\times1-5\times(-1) & 2\times3-3\times(-3)-5\times3 & 2\times5-3\times(-5)-5\times3 \\ -1\times(-1)+4\times1+5\times(-1) & -1\times3+4\times(-3)+5\times3 & -1\times5+4\times(-5)+5\times3 \\ 1\times(-1)-3\times1-4\times(-1) & 1\times3-3\times(-3)-4\times3 & 1\times5-3\times(-5)-4\times3 \end{bmatrix}$$

$$= \begin{bmatrix} -2-3+5 & 6+8-15 & 10+15-15 \\ 1+4-5 & -3-12+15 & -5-20+15 \\ -1-3+5 & 3+9-12 & 5+15-12 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 10 \\ 0 & 0 & -10 \\ 0 & 0 & 8 \end{bmatrix}.$$

1.(vi) BA

Sol.

$$BA = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 3 \end{bmatrix}\begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} -1\times2+3\times(-1)+5\times1 & -1\times(-3)+3\times4+5\times(-3) & -1\times(-5)+3\times5+5\times(-4) \\ 1\times2-3\times(-1)-5\times1 & 1\times(-3)-3\times4-5\times(-3) & 1\times(-5)-3\times5-5\times(-4) \\ -1\times2+3\times(-1)+3\times1 & -1\times(-3)+3\times4+3\times(-3) & -1\times(-5)+3\times5+3\times(-4) \end{bmatrix}$$

$$= \begin{bmatrix} -2-3+5 & 3+12-15 & 5+15-20 \\ 2+3-5 & -3-12+15 & -5-15+20 \\ -2-3+3 & 3+12-9 & 5+15-12 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -2 & 6 & 8 \end{bmatrix}.$$

2. (i) Evaluate $\begin{bmatrix} 1 & 2 & 1 \\ 4 & 0 & 2 \end{bmatrix}\begin{bmatrix} 2 & -4 \\ 1 & 5 \\ -2 & 2 \end{bmatrix}$

Sol. $\begin{bmatrix} 1 & 2 & 1 \\ 4 & 0 & 2 \end{bmatrix}\begin{bmatrix} 2 & -4 \\ 1 & 5 \\ -2 & 2 \end{bmatrix}$

$$= \begin{bmatrix} 2+2-2 & -4+10+2 \\ 8+0-4 & -16+0+4 \end{bmatrix} = \begin{bmatrix} 2 & 8 \\ 4 & -12 \end{bmatrix}$$

2.(ii) $[x\ y\ z]\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Sol. $[x\ y\ z]\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = [x\ y\ z]\begin{bmatrix} ax+hy+gz \\ hx+by+fz \\ gx+fy+cz \end{bmatrix}$

$$= [x(ax+hy+gz)+y(hx+by+fz)+z(gx+fy+cz)]$$

$$= [ax^2+hxy+gzx+hxy+by^2+fyz+gzx+fyz+cz^2]$$

$$= [ax^2+by^2+cz^3+2hxy+2fyz+2gzx]$$

2.(iii) $\begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^2$

Sol. $\begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^2 = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$

$$= \begin{bmatrix} 1+8 & 2-6 \\ 4-12 & 8+9 \end{bmatrix} = \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix}$$

2.(iv) $\begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^3$

Sol. $\begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^3 = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}^2 \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$

$= \begin{bmatrix} 9 & -4 \\ -8 & 17 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$, From 2(*iii*) above

$= \begin{bmatrix} 9-16 & 18+12 \\ -8+68 & -16-51 \end{bmatrix} = \begin{bmatrix} -7 & 30 \\ 60 & -67 \end{bmatrix}$

2.(v) $\begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^2$

Sol. $\begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^2 = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1-0-0 & 1-1-0 & 1-0-1 \\ 0+0+0 & 0+1+0 & 0+0+0 \\ 0+0+0 & 0+0+0 & 0+0+1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

2.(vi) $\begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^3$

Sol. $\begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^3 = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^2 \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

by 2(v) above

$= \begin{bmatrix} -1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

3. Prove that the product of matrices

$$A=\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix} \text{ and } B=\begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$$

is the zero matrix when θ and ϕ differ by an odd multiple of $\frac{\pi}{2}$.

Sol. $$AB = \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix}\begin{bmatrix} \cos^2\phi & \cos\phi\sin\phi \\ \cos\phi\sin\phi & \sin^2\phi \end{bmatrix}$$

$$=\begin{bmatrix} \cos^2\theta\cos^2\phi+\cos\theta\sin\theta\cos\phi\sin\phi & \cos^2\theta\cos\phi\sin\phi+\cos\theta\sin\theta\sin^2\phi \\ \cos\theta\sin\theta\cos^2\phi+\sin^2\theta\cos\phi\sin\phi & \cos\theta\sin\theta\cos\phi\sin\phi+\sin^2\theta\sin^2\phi \end{bmatrix}$$

$$=\begin{bmatrix} \cos\theta\cos\phi(\cos\theta\cos\phi+\sin\theta\sin\phi) & \cos\theta\sin\phi(\cos\theta\cos\phi+\sin\theta\sin\phi) \\ \sin\theta\cos\phi(\cos\theta\cos\phi+\sin\theta\sin\phi) & \sin\theta\sin\phi(\cos\theta\cos\phi+\sin\theta\sin\phi) \end{bmatrix}$$

$$=\begin{bmatrix} \cos\theta\cos\phi\cos(\theta-\phi) & \cos\theta\sin\phi\cos(\theta-\phi) \\ \sin\theta\cos\phi\cos(\theta-\phi) & \sin\theta\sin\phi\cos(\theta-\phi) \end{bmatrix}$$

$$=\cos(\theta-\phi)\begin{bmatrix} \cos\theta\cos\phi & \cos\theta\sin\phi \\ \sin\theta\cos\phi & \sin\theta\sin\phi \end{bmatrix} = 0, \text{ if and only if } \cos(\theta-\phi)=0.$$

Hence $AB = 0, \quad \Leftrightarrow \quad \cos(\theta-\phi)=0$

or $AB = 0, \quad \Leftrightarrow \quad \theta-\phi = (2n+1)\frac{\pi}{2}$, where n is an integer.

4. The direction cosines of two lines are $\lambda_1, \lambda_2, \lambda_3$ and μ_1, μ_2, μ_3. Prove that the product

$$\begin{bmatrix} \lambda_1^2 & \lambda_1\lambda_2 & \lambda_1\lambda_3 \\ \lambda_1\lambda_2 & \lambda_2^2 & \lambda_2\lambda_3 \\ \lambda_1\lambda_3 & \lambda_2\lambda_3 & \lambda_3^2 \end{bmatrix}\begin{bmatrix} \mu_1^2 & \mu_1\mu_2 & \mu_1\mu_3 \\ \mu_1\mu_2 & \mu_2^2 & \mu_2\mu_3 \\ \mu_1\mu_3 & \mu_2\mu_3 & \mu_3^2 \end{bmatrix}$$

is zero if and only if the lines are perpendicular to each other.

Sol. $$AB = \begin{bmatrix} \lambda_1^2 & \lambda_1\lambda_2 & \lambda_1\lambda_3 \\ \lambda_1\lambda_2 & \lambda_2^2 & \lambda_2\lambda_3 \\ \lambda_1\lambda_3 & \lambda_2\lambda_3 & \lambda_3^2 \end{bmatrix}\begin{bmatrix} \mu_1^2 & \mu_1\mu_2 & \mu_1\mu_3 \\ \mu_1\mu_2 & \mu_2^2 & \mu_2\mu_3 \\ \mu_1\mu_3 & \mu_2\mu_3 & \mu_3^2 \end{bmatrix}$$

$$=\begin{bmatrix} \lambda_1^2\mu_1^2+\lambda_1\lambda_2\mu_1\mu_2+\lambda_1\lambda_3\mu_1\mu_3 & \lambda_1^2\mu_1\mu_2+\lambda_1\lambda_2\mu_2^2+\lambda_1\lambda_3\mu_2\mu_3 & \lambda_1^2\mu_1\mu_3+\lambda_1\lambda_2\mu_2\mu_3+\lambda_1\lambda_3\mu_3^2 \\ \lambda_1\lambda_2\mu_1^2+\lambda_2^2\mu_1\mu_2+\lambda_2\lambda_3\mu_1\mu_3 & \lambda_1\lambda_2\mu_1\mu_2+\lambda_2^2\mu_2^2+\lambda_2\lambda_3\mu_2\mu_3 & \lambda_1\lambda_2\mu_1\mu_3+\lambda_2^2\mu_2\mu_3+\lambda_2\lambda_3\mu_3^2 \\ \lambda_1\lambda_3\mu_1^2+\lambda_2\lambda_3\mu_1\mu_2+\lambda_3^2\mu_1\mu_3 & \lambda_1\lambda_3\mu_1\mu_2+\lambda_2\lambda_3\mu_2^2+\lambda_3^2\mu_2\mu_3 & \lambda_1\lambda_3\mu_1\mu_3+\lambda_2\lambda_3\mu_2\mu_3+\lambda_3^2\mu_3^2 \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_1\mu_1(\lambda_1\mu_1+\lambda_2\mu_2+\lambda_3\mu_3) & \lambda_1\mu_2(\lambda_1\mu_1+\lambda_2\mu_2+\lambda_3\mu_3) & \lambda_1\mu_3(\lambda_1\mu_1+\lambda_2\mu_2+\lambda_3\mu_3) \\ \lambda_2\mu_1(\lambda_1\mu_1+\lambda_2\mu_2+\lambda_3\mu_3) & \lambda_2\mu_2(\lambda_1\mu_1+\lambda_2\mu_2+\lambda_3\mu_3) & \lambda_2\mu_3(\lambda_1\mu_1+\lambda_2\mu_2+\lambda_3\mu_3) \\ \lambda_3\mu_1(\lambda_1\mu_1+\lambda_2\mu_2+\lambda_3\mu_3) & \lambda_3\mu_2(\lambda_1\mu_1+\lambda_2\mu_2+\lambda_3\mu_3) & \lambda_3\mu_3(\lambda_1\mu_1+\lambda_2\mu_2+\lambda_3\mu_3) \end{bmatrix}$$

$$= (\lambda_1\mu_1 + \lambda_2\mu_2 + \lambda_3\mu_3) \begin{bmatrix} \lambda_1\mu_1 & \lambda_1\mu_2 & \lambda_1\mu_3 \\ \lambda_2\mu_1 & \lambda_2\mu_2 & \lambda_2\mu_3 \\ \lambda_3\mu_1 & \lambda_3\mu_2 & \lambda_3\mu_3 \end{bmatrix}$$

Hence $AB = 0$ if and only if $\lambda_1\mu_1 + \lambda_2\mu_2 + \lambda_3\mu_3 = 0$.

i.e., if and only if the two lines are perpendicular.

5. Show that, in general, for any two matrices A and B which are conformable for addition and multiplication,

$$(A+B)^2 \neq A^2 + 2AB + B^2$$

and $$A^2 - B^2 \neq (A-B)(A+B)$$

Under what conditions equality holds in each case?

Sol. $(A+B)^2 = (A+B(A+B) = A^2 + AB + BA + B^2$

Since $AB \neq BA$ in general, so we must have

$$(A+B)^2 \neq A^2 + 2AB + B^2$$

$$(A-B)(A+B) = A^2 + AB - BA - B^2 \neq A^2 - B^2.$$

since $AB \neq BA$ is general.

Equality holds in each case if $AB = BA$.

6. If

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}, \text{ show that}$$

$$A^2 - 4A - 5I = \begin{bmatrix} 2 & -2 & 2 \\ 0 & 0 & -2 \\ 0 & 0 & -2 \end{bmatrix} \text{ and}$$

$$A^3 - 3A^2 - 7A - 3I = 0.$$

Sol. $A^2 - 4A - 5I$

$$= \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 4\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 5\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+4+2 & 2+2+2 & 1+4+1 \\ 2+2+4 & 4+1+4 & 2+2+2 \\ 2+4+2 & 4+2+2 & 2+4+1 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 4 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 6 & 6 \\ 8 & 9 & 6 \\ 8 & 8 & 7 \end{bmatrix} + \begin{bmatrix} -9 & -8 & -4 \\ -8 & -9 & -8 \\ -8 & -8 & -9 \end{bmatrix} = \begin{bmatrix} -2 & -2 & 2 \\ 0 & 0 & -2 \\ 0 & 0 & -2 \end{bmatrix}$$

$$A^3 - 3A^2 - 7A - 3I = A\,.\,A^2 - 3A^2 - 7A - 3I$$

$$= \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}\begin{bmatrix} 7 & 6 & 6 \\ 8 & 9 & 6 \\ 8 & 8 & 7 \end{bmatrix} - 3\begin{bmatrix} 7 & 6 & 6 \\ 8 & 9 & 6 \\ 8 & 8 & 7 \end{bmatrix} - 7\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 3\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 7+16+8 & 6+18+8 & 6+12+7 \\ 14+8+16 & 12+9+16 & 12+6+14 \\ 14+16+8 & 12+18+8 & 12+12+7 \end{bmatrix} - \begin{bmatrix} 21 & 18 & 18 \\ 24 & 27 & 18 \\ 24 & 24 & 21 \end{bmatrix}$$

$$- \begin{bmatrix} 7 & 14 & 7 \\ 14 & 7 & 14 \\ 14 & 14 & 7 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 31 & 32 & 25 \\ 38 & 37 & 32 \\ 38 & 38 & 31 \end{bmatrix} + \begin{bmatrix} -21-7-3 & -18-14 & -18-7 \\ -24-14 & -27-7-3 & -18-14 \\ -24-14 & -24-14 & -21-7-3 \end{bmatrix}$$

$$= \begin{bmatrix} 31 & 32 & 25 \\ 38 & 37 & 32 \\ 38 & 38 & 31 \end{bmatrix} + \begin{bmatrix} -31 & -32 & -25 \\ -38 & -37 & -32 \\ -38 & -38 & -31 \end{bmatrix} = O.$$

7. Show that the matrix

$$A = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} \text{ is periodic having period 2.}$$

Sol. $$A^2 = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}\begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1+6-12 & -2-4+0 & -6-18+18 \\ -3-6+18 & 6+4+0 & 18+18-27 \\ 2+0-6 & -4+0+0 & -12+0+9 \end{bmatrix} = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix}\begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -5+18-12 & 10-12+0 & 30-54+18 \\ 9-30+18 & -18+20+0 & -54+90-27 \\ -4+12-6 & 8-8+0 & 24-36+9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} = A$$

Thus A is periodic having period 2.

8. Show that

$$\begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix} \text{ is nilpotent. What is its nilpotency index?}$$

Sol. Let $A = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}\begin{bmatrix} 1 & -3 & -4 \\ -1 & 3 & 4 \\ 1 & -3 & -4 \end{bmatrix}$$

$$= \begin{bmatrix} 1+3-4 & -3-9+12 & -4-12+16 \\ -1-3+4 & 3+9-12 & 4+12-16 \\ 1+3-4 & -3-9+12 & -4-12+16 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O.$$

Hence A is nilpotent of index 2.

9. Show that

$$\begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix} \text{ is involutory.}$$

Sol. Let $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$. Then

$$A^2 = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}\begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0+4-3 & 0-3+3 & 0+4-4 \\ 0-12+12 & 4+9-12 & -4-12+16 \\ 0-12+12 & 3+9-12 & -3-12+16 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$